

Flaring in the Permian: what satellites see and what operators report

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GitHub Repository: <https://github.com/Erdos-Projects/summer26-permian-flaring>

Project Objective: Texas operators self report vented and flared gas volumes to the Railroad Commission (RRC); that series lags and stays preliminary for months. VIIRS Nightfire observes the same flares nightly, which is independent of any oil/gas production data. We ask two questions. Can satellite radiant heat estimate flared volume ahead of the official release, relative to a baseline? And since flared gas is a fraction of casinghead gas, which ties to oil through the gas-oil ratio, can that chain be inverted to estimate oil production? If not, which link is unstable?

Data Preparation: VIIRS Nightfire per-site radiant heat (EOG, 2012–2025) is filtered to the Permian, aggregated monthly, normalized per clear look, and masked for the August 2022 satellite outage. RRC lease-level production and disposition data are joined to well lists and coordinates, with lease volumes allocated uniformly across wells and wells assigned to the Nightfire site containing them. After this processing we get a site-month panel of paired flaring and production. This methodology has two limitations: we are assuming uniform allocation and, the fact that most basin wells fall outside any flare site, so only a minority of basin production maps onto the satellite footprint.

Results : Forecasting Flared Volume: An ARIMAX(1,1,1) using radiant heat (per clear look) as an exogenous regressor is run in two modes: a nowcast of the current month’s flared volume from current satellite data, weeks ahead of the RRC print, and a one step ahead forecast. Both were validated by 92-month walk forward backtest (2018–2025) against the final revised RRC volume. The satellite regressor lifts mean directional accuracy from 56.5% (satellite-free ARIMA) to 68.5%. OLS on the same regressor matches on direction but more than doubles absolute error, lacking an autoregressive term to anchor levels; XGBoost tracks direction (65.2%) but cannot extrapolate past its training range and degrades on the largest surges.

Results - The Chain to Oil: The second question has a clean negative answer. In the chain VIIRS Radiant Heat → flared gas → total casinghead gas → oil, the last two links are unstable across our window: the flared fraction falls significantly to its current level of roughly 0.02 as major pipelines come online (2019-2020), and the gas-oil ratio more than doubles. Compounding this, flared gas is only ~2% of casinghead gas, so inverting the chain divides by a small number and amplifies any error in the flaring estimate by a large factor. Through 2019 oil and radiant heat rise together; after 2019 oil is flat while radiant heat falls sharply and models targeting oil do not beat a persistence baseline.

Executive Impact: Flared volume can be estimated from satellite data before the RRC releases it, with directional accuracy well above a satellite free baseline. Flaring surges signal that the regional gas takeaway process is at capacity, thus an early read on flaring is useful to commodity analysts and to midstream planners. For regulators, the value is exogeneity: VIIRS does not depend on operator filings. Two limits: the model is basin aggregate and monthly, so it does not identify individual sites; and since it is calibrated on reported volumes, it can catch a change in reporting, not a persistent bias in it.

Future Directions: If the signal measures decoupling, decoupling should be the object of study. New Mexico's 2021 venting and flaring rule imposes a 98% capture requirement; Texas has no equivalent; and the Delaware sub-basin straddles the state line with shared geology, operators, and pipelines. That supports a difference-in-differences design on the flared fraction of casinghead gas, with pipeline in service dates as a falsification test : they should move both states, while the rule should move only New Mexico. Preliminary structural break analysis on the Texas side is consistent with this: breaks align to pipeline commissioning, not to the rule date.